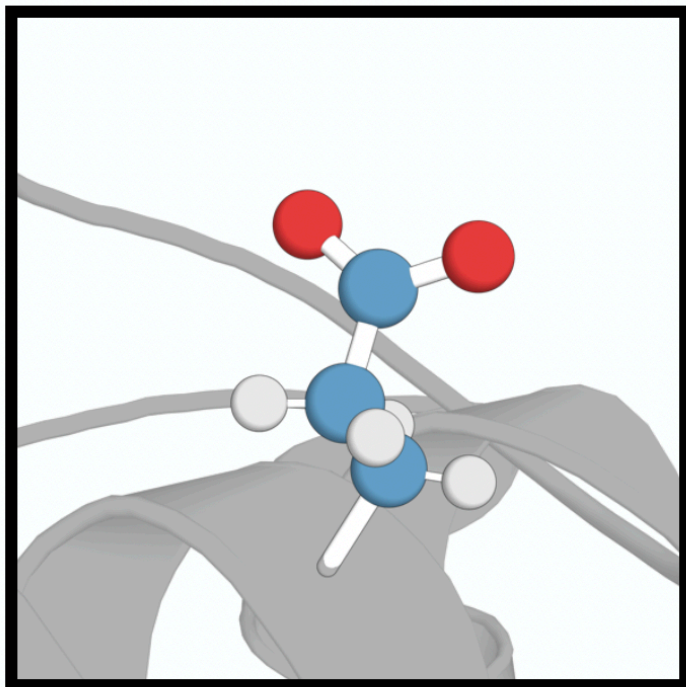
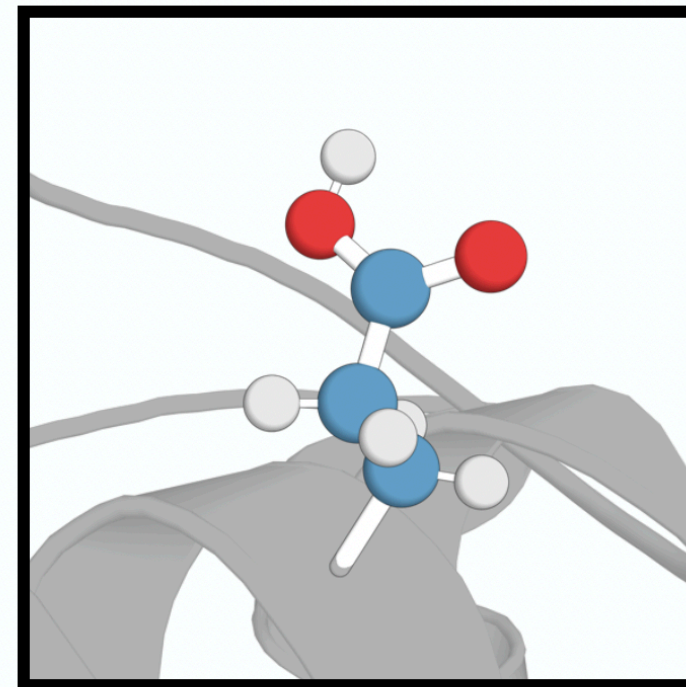




$$\Delta G_{\text{eff}}$$

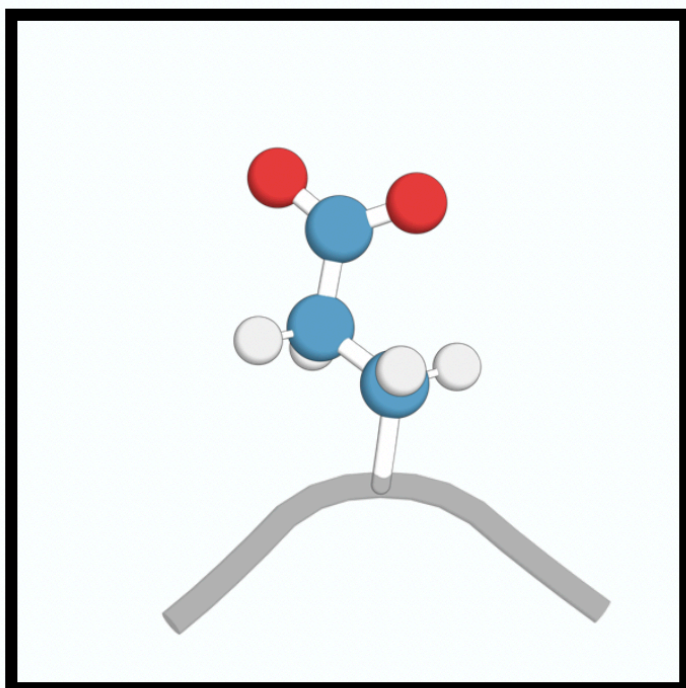
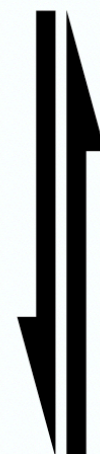


$$\Delta G_{\text{deprotonated}}^{\text{transfer}}$$



$$\Delta G_{\text{deprotonated}}^{\text{transfer}} - \Delta G_{\text{protonated}}^{\text{transfer}} = \Delta U + \Delta G_{\text{A}^-}^{\text{ref}}$$

$$\Delta G_{\text{protonated}}^{\text{transfer}}$$



$$k_B T \ln 10 (\text{pH} - \text{p}K_a^{\text{ref}})$$

